

Mathematical Model of MTPA Control for M- and T-axis Current Control Systems in PMSM Drives

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(Manuscript received May 10, 2024, revised Oct. 21, 2024)
J-STAGE Advance published date : Feb. 21, 2025

This paper proposes a relationship between the M- and T-axis currents for the maximum torque per ampere (MTPA) control of permanent magnet synchronous motors (PMSMs). This paper focuses on a rotating reference frame, which is synchronized with the stator flux-linkage vector. The validity of the proposed relationship is confirmed by using finite element analysis (FEA) and measured values of the tested motor. The control characteristics of the M- and T-axis current control system are also evaluated.

Keywords: M-T frame, maximum torque per ampere (MTPA), M- and T-axis current, permanent magnet synchronous motor (PMSM), stator flux linkage

1. Introduction

Permanent magnet synchronous motors (PMSMs) are used as high efficiency motors in industrial application or home appliances. In PMSM drives, the d - q frame, which is synchronized with the rotor position, is generally used⁽¹⁾⁽²⁾. In the d - q frame, maximum torque per ampere (MTPA) control of PMSM is available as a highly efficient control. However, when high power and torque density are required, for short it is necessary to increase torque or to reduce size of motor, the magnetic saturation or cross-coupling effects are significant and the motor parameters vary according to the current. The parameter variation affects the performance of PMSM control. Therefore, highly efficient control in d - q frame is not easy when the torque or power density are raised. For this measure, the methods which resolve these problems have been considered. For instance, the models of d - and q -axis inductances which are affected by d - and q -axis currents have been detailed or identified as online parameters⁽³⁾⁽⁴⁾. However, the accurate models tend to be complicated. Therefore, the easier methods have been required.

M-T frame that is synchronized with stator flux linkage can be also used for synchronous machine drives^{(5)–(9)}. In this frame, the model of torque and voltage equation are expressed simply. Thus, M-T frame has been applied direct torque control (DTC) and direct flux vector control because it is easy to construct controller and to calculate torque^{(10)–(15)}. It has been known that there is a quadratic function-like

relationship between M- and T-axis currents under the MTPA condition⁽¹⁶⁾. However, the relational expression has not been clearly studied. In the d - and q -axis current control, motor parameters such as the d - and q -axis inductances, and the magnet flux are used for MTPA control. In contrast, an approach based on the M-T frame aims to simplify the controller by using a single constant for MTPA. Thus, clarifying the relationship between M- and T-axis currents under MTPA conditions is essential.

This study focuses on modeling the relationship between the M- and T-axis currents as a mathematical equation⁽¹⁷⁾. The relationship between the M- and T-axis currents for the MTPA control of PMSM is expressed as a quadratic function. This paper proposes a mathematical model of the MTPA control used for the M- and T-axis current control system. The coefficient used for the proposed model can be calculated from well-known machine parameters, such as the d - and q -axis inductances and the magnet flux. The validity of the proposed model is confirmed by using FEA and experiment. This paper also discusses M- and T-axis current control system.

2. PMSM Model in M-T Frame

2.1 Definition of M-T Frame Fig. 1 shows a vector diagram of PMSM in the d - q and the M-T frames⁽¹⁾. ψ_a is the stator flux linkage vector due to permanent magnets, ψ_o is the stator flux linkage vector, δ_o is the torque angle which is the angle between ψ_a and ψ_o , i_a is an armature current vector, and i_d , i_q , i_M , and i_T are the d -, q -, M-, and T-axis components of i_a .

i_M and i_T are expressed as

$$\begin{bmatrix} i_M \\ i_T \end{bmatrix} = \begin{bmatrix} \cos \delta_o & \sin \delta_o \\ -\sin \delta_o & \cos \delta_o \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \dots \dots \dots (1)$$

i_M and i_T are obtained from i_d and i_q by coordinate transformation using δ_o which is defined as

This paper is based on Reference (17), which published in the 23rd International Conference on Electrical Machines and Systems (ICEMS2020) ©2020 IEEJ.

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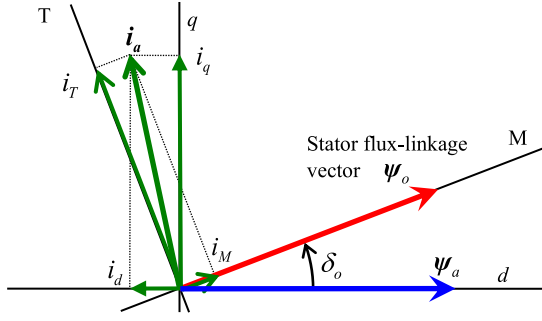


Fig. 1. Vector diagram in the M-T frame

$$\delta_o = \tan^{-1} \left(\frac{L_q i_q}{\Psi_a + L_d i_d} \right) \dots \dots \dots (2)$$

where Ψ_a is stator flux linkage due to permanent magnets, L_d and L_q are d - and q -axis inductances respectively.

The voltage equation in M-T frame is expressed as

$$\begin{bmatrix} v_M \\ v_T \end{bmatrix} = R_a \begin{bmatrix} i_M \\ i_T \end{bmatrix} + \begin{bmatrix} d \\ dt \end{bmatrix} \Psi_o \dots \dots \dots (3)$$

where R_a is armature resistance, ω is electrical angular frequency, and Ψ_o is stator flux linkage.

2.2 Mathematical Model of MTPA Curve Considering the mathematical model of the MTPA curve of PMSM in the Ψ_o - i_T plane. This MTPA curve is applied to DTC in M-T frame. It can be modeled as

$$\Psi_o = g(i_T) i_T + \Psi_a \dots \dots \dots (4)$$

where g is the variable which decides form of the curve.

The MTPA curve of PMSM without considering the effect of magnetic saturation is expressed as⁽⁵⁾

$$\Psi_o = L_T i_T \frac{2}{\pi} \tan^{-1} \left(\frac{L_k}{\Psi_a} i_T \right) + \Psi_a \dots \dots \dots (5)$$

where L_T and L_k are the variables which decides form of the curve.

In (5), the term L_k/Ψ_a in the arctangent is a factor that influences the shape of MTPA curves. The obtained value is normalized by multiplying by $2/\pi$ due to the arctangent's asymptotic limit of $\pi/2$. The value of L_T , corresponding to inductance, corresponds to the slope of the MTPA curve when the current i_T is sufficiently large.

Equation (5), involving the arctangent function, necessitates a lengthy computation time to determine the relationship between the stator flux linkage and the T-axis current in a microprocessor-based control system. A simplified MTPA model is therefore derived from (5). When L_k approaches zero in (5), the simplified model is given by⁽¹⁵⁾

$$\Psi_o = k_{\Psi_o-i_T} i_T^2 + \Psi_a \dots \dots \dots (6)$$

where $k_{\Psi_o-i_T}$ is the variable which decides form of the curve.

Ψ_o can be expressed as a quadratic function of i_T and this model can achieve reduced computational complexity through the use of a sum of products operation.

Although Ψ_o and i_T are orthogonal components in the M-T frame and should be unrelated, the MTPA curve can be expressed in terms of the relationship between Ψ_o and i_T .

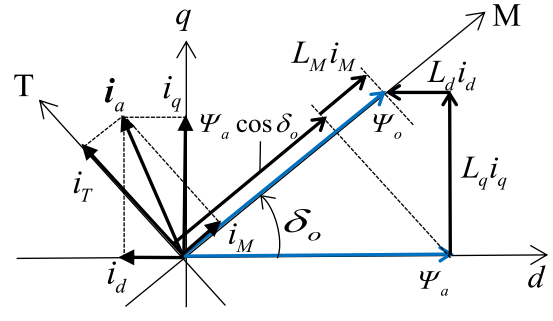


Fig. 2. Definition of M-axis inductance

In addition, although the methods for determining the optimal values of constants such as L_T , L_k and $k_{\Psi_o-i_T}$ have been established, the relationship between these constants and the well-known machine parameters, such as Ψ_a , L_d and L_q , have still to be determined.

2.3 Relationship between Ψ_o and i_M in PMSM

Fig. 2 shows the definition of M-axis inductance in PMSM. From Fig. 2, Ψ_o is expressed as

$$\Psi_o = L_M i_M + \Psi_a \cos \delta_o \dots \dots \dots (7)$$

where L_M is defined as M-axis inductance in this study.

Assuming that Ψ_o can be approximated by a linear equation for i_M , Ψ_o is expressed as

$$\Psi_o = k_{\Psi_o-i_M} i_M + \Psi_a \dots \dots \dots (8)$$

where $k_{\Psi_o-i_M}$ is coefficient of linear function.

Consider expressing $k_{\Psi_o-i_M}$ as a motor parameter. $k_{\Psi_o-i_M}$ can be written as

$$k_{\Psi_o-i_M} = \frac{\Psi_o - \Psi_a}{i_M} \dots \dots \dots (9)$$

With machine parameters, Ψ_o is written as

$$\Psi_o = \sqrt{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2} \dots \dots \dots (10)$$

With (1), (9) and (10), $k_{\Psi_o-i_M}$ is written as

$$k_{\Psi_o-i_M} = \frac{\sqrt{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2} - \Psi_a}{i_d \cos \delta_o + i_q \sin \delta_o} \dots \dots \dots (11)$$

When using machine parameters, $\sin \delta_o$ and $\cos \delta_o$ are expressed as

$$\sin \delta_o = \sin \left(\tan^{-1} \left(\frac{L_q i_q}{\Psi_a + L_d i_d} \right) \right) = \frac{L_q i_q}{\sqrt{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2}} \dots \dots \dots (12)$$

$$\cos \delta_o = \cos \left(\tan^{-1} \left(\frac{L_q i_q}{\Psi_a + L_d i_d} \right) \right) = \frac{\Psi_a + L_d i_d}{\sqrt{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2}} \dots \dots \dots (13)$$

Therefore, $k_{\Psi_o-i_M}$ can be expressed as

$$k_{\Psi_o-i_M} = \frac{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2 - \Psi_a \sqrt{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2}}{(\Psi_a + L_d i_d) i_d + L_q i_q \cdot i_q} \dots \dots \dots (14)$$

$k_{\Psi_o-i_M}$ can be calculated by giving L_d , L_q , and Ψ_a as constants, since it includes parameters such as L_d , L_q , and Ψ_a ,

which are often expressed as

$$L_M = \frac{\Psi_a(1 - \cos \delta_o) \sqrt{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2}}{(\Psi_a + L_d i_d) i_d + L_q i_q \cdot i_q} + k_{\Psi_o-i_M} \quad (15)$$

2.4 Relationship between i_M and i_T in PMSM It has been known that there is a quadratic function-like relationship between i_M and i_T under the MTPA condition⁽¹⁶⁾. In this paper, relationship between i_M and i_T is derived by an equation using known motor parameters. Assuming the relationship between i_M and i_T is a quadratic function, it is expressed as

$$i_M = k_{i_M-i_T} i_T^2 \quad (16)$$

where $k_{i_M-i_T}$ is a constant.

Considering to express $k_{i_M-i_T}$ using the parameters of the motor model in the well-known d - q frame. The following equation is derived from (16).

$$k_{i_M-i_T} = i_M / i_T^2 \quad (17)$$

Although L_d , L_q , and Ψ_a in the actual motor have the characteristics of varying with the current, we can consider δ_o as a function of the current only by treating them as constants. By applying this δ_o to (1), this varies due to currents only regardless of other parameters. Thus, (1) can be expressed as a function of currents only same as δ_o . By substituting the obtained i_M and i_T into (17), $k_{i_M-i_T}$ is expressed as

$$k_{i_M-i_T} = \frac{\{(\Psi_a + L_d i_d) i_d + L_q i_q \cdot i_q\} \sqrt{(\Psi_a + L_d i_d)^2 + (L_q i_q)^2}}{\{-L_q i_q i_d + (\Psi_a + L_d i_d) i_q\}^2} \quad (18)$$

3. Validity of Proposed Relationship

3.1 Relationship between Ψ_o and i_M Table 1 shows machine parameters of surface permanent-magnet synchronous motor (SPMSM) and interior permanent-magnet synchronous motor (IPMSM) used for the evaluation. The current characteristics were obtained by using FEA. An FEA software developed in our laboratory was used in this study. L_d and L_q are the values at the rated current and Ψ_a is the value at the current of 0 A in Table 1.

Fig. 3 shows the characteristics of Ψ_o and i_M for SPMSM and IPMSM. These characteristics are calculated using (14) from current characteristics obtained by the FEA. From Fig. 3, it can be seen that although the current is included as a variable, the effect of the current change on $k_{\Psi_o-i_M}$ is small for both SPMSM and IPMSM. Therefore, the Ψ_o - i_M characteristics can be written by using $k_{\Psi_o-i_M}$ at the rated current.

Fig. 4 shows a comparison of the characteristics obtained by FEA with those obtained using $k_{\Psi_o-i_M}$ at rated current. This $k_{\Psi_o-i_M}$ is the value obtained with the motor parameter given as a constant. The characteristics obtained by FEA is that in the case of SPMSM, Ψ_o varies linearly with respect to i_M . In the case of IPMSM, the linearity is lost with increasing current value due to magnetic saturation, but it can be regarded as a linear approximation up to the rated current. In both cases, the Ψ_o - i_M characteristics can be obtained up to the vicinity of the rated current.

Table 1. Machine Parameters

	SPMSM	IPMSM
Number of pole pairs P_n	2	
d -axis inductance L_d	5.06 mH	6.37 mH
q -axis inductance L_q	5.13 mH	15.5 mH
Permanent magnet flux linkage Ψ_a	0.113 Wb	0.0861 Wb
Armature resistance R_a	0.814 Ω	
Rated current I_{am}	6.79 A	

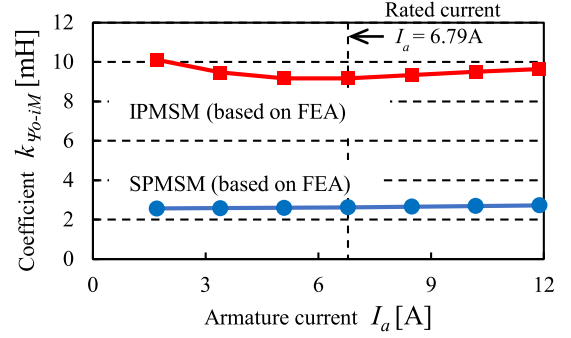


Fig. 3. Coefficient of stator flux linkage for armature current, calculated by (14)

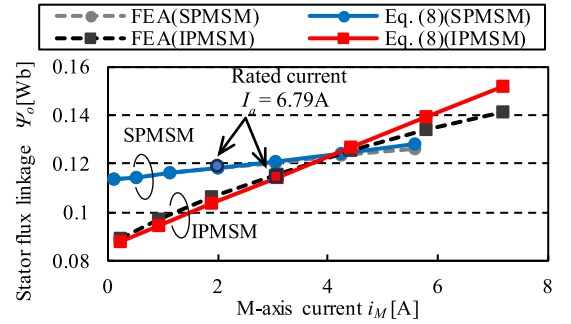


Fig. 4. Characteristics of stator flux linkage for M-axis current

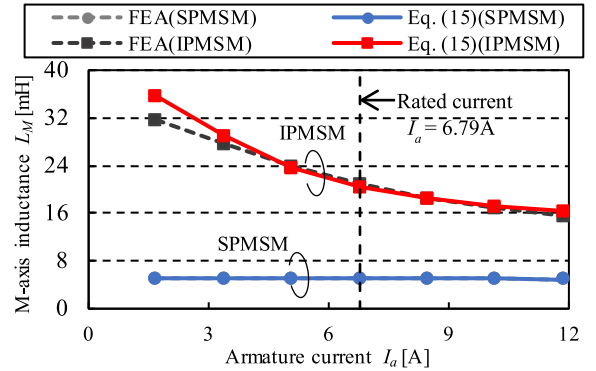


Fig. 5. Comparison of L_M by FEA and (15)

Fig. 5 shows a comparison of the L_M obtained by (15) and the FEA. From Fig. 5, the results obtained by (15) and FEA are generally similar.

Fig. 6 shows a comparison of the characteristics of δ_o . This shows that there is not much difference between the results obtained by FEA and the results obtained by the constant parameters are calculated. Thus, by using δ_o which is calculated with parameters, (12) and (13) can be regarded as function of currents. Accordingly, (14) and (15) are expressed as

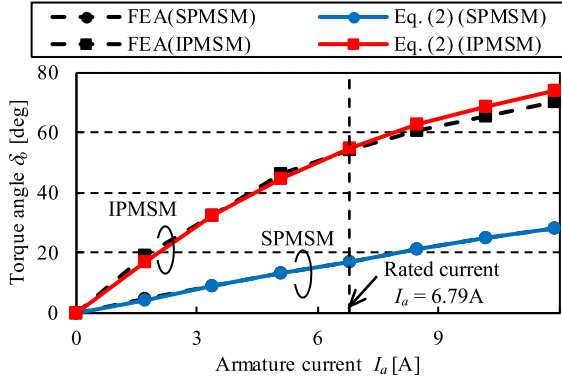


Fig. 6. Torque angle characteristics under MTPA condition

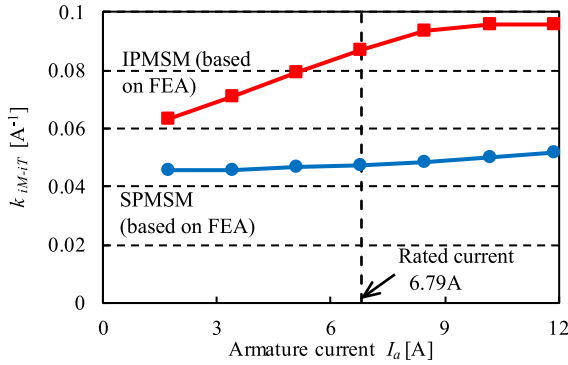


Fig. 7. Characteristics of (18)

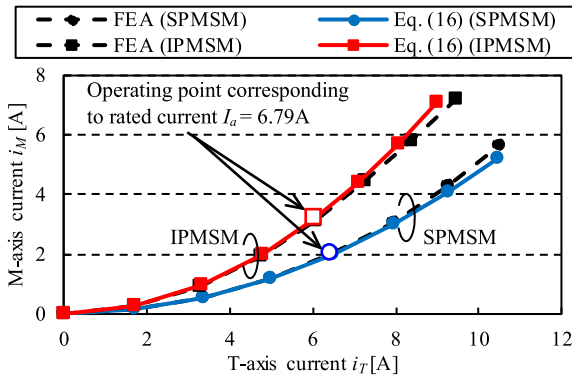


Fig. 8. M- and T-axis currents characteristics under MTPA condition

function of currents.

3.2 Relationship between i_M and i_T Fig. 7 shows the characteristics of k_{iM-iT} given by (18). Here, the value of (18) is calculated from current characteristics obtained by the FEA. In Fig. 7, the characteristic of the SPMSM is almost constant, and the IPMSM has a larger variation than the SPMSM, but this is because the IPMSM has a larger variation of inductance due to magnetic saturation than the SPMSM.

Fig. 8 shows the comparison between the FEA of the i_M and i_T and the approximate equation in (16), where k_{iM-iT} is the result of (18) where the rated current is substituted for k_{iM-iT} (SPMSM: $k_{iM-iT} = 0.0474 \text{ A}^{-1}$, IPMSM: $k_{iM-iT} = 0.0873 \text{ A}^{-1}$). In addition, i_T is calculated by substituting the parameters in Table 1 and the d - and q -axis currents at the rated current. It can be seen that the approximation in (16) is generally successful for both SPMSM and IPMSM below the rated current value. These results confirm that the relationship of the M- and T-axis currents in the PMSM can be expressed as a quadratic function using the motor parameters.

4. Current Vector Control in M-T Frame

4.1 Configuration of Current Controller Fig. 9 shows the scheme of M- and T-axis currents control system. Reference M- and T-axis currents i_M^* and i_T^* are calculated by (16). The M- and T-axis current controllers are based on a proportional and integral (PI) control. Controlling obtained deviation of reference and actual values of M- and T-axis currents, reference values of M- and T-axis voltages are generated.

In the case of coordinate transformation between d - and q -axis components and three phase components, rotor position θ is used. On the other hand, coordinate transformation between M- and T-axis components and three phase components needs stator flux linkage position θ_o . θ_o is the angle which is sum of rotor position and torque angle. In Fig. 9, rotor position θ and torque angle δ_o are used to calculate stator flux linkage position θ_o . In the torque angle calculator, δ_o is obtained by (2), and thus, the machine parameters are used.

4.2 Simulation Results Simulation results of M- and T-axis current vector control system for SPMSM is shown as Fig. 10 and another for IPMSM is shown as Fig. 11. In this simulation, the reference current is changed from 0 A to the

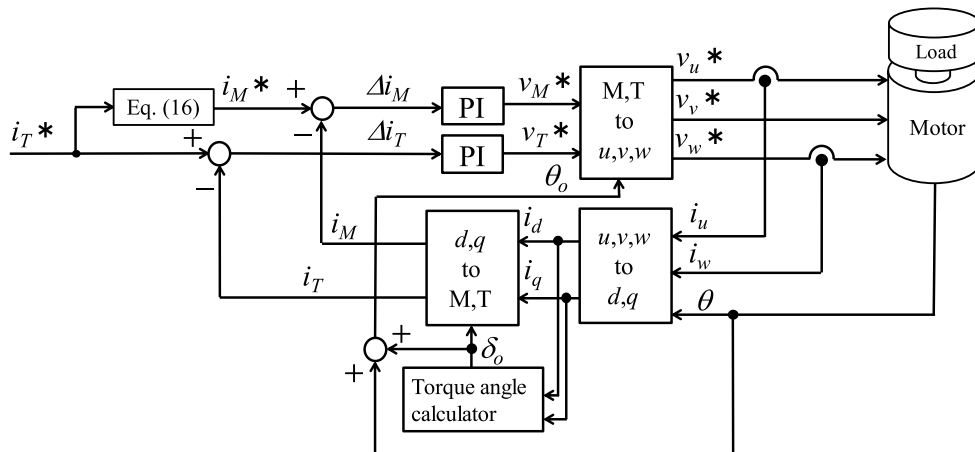
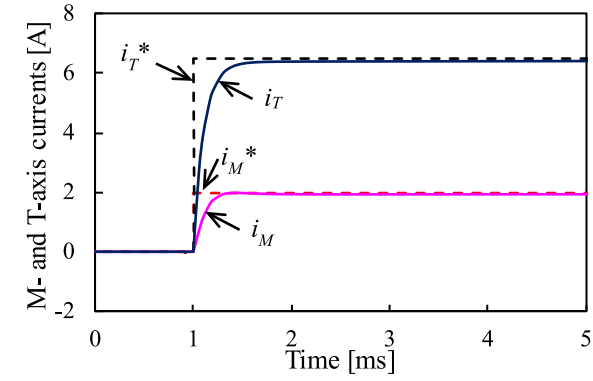
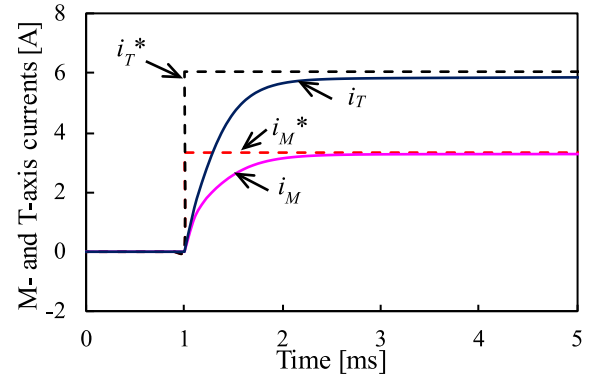


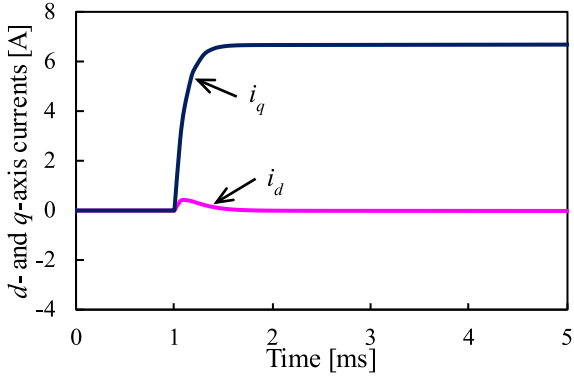
Fig. 9. M- and T-axis currents vector control system



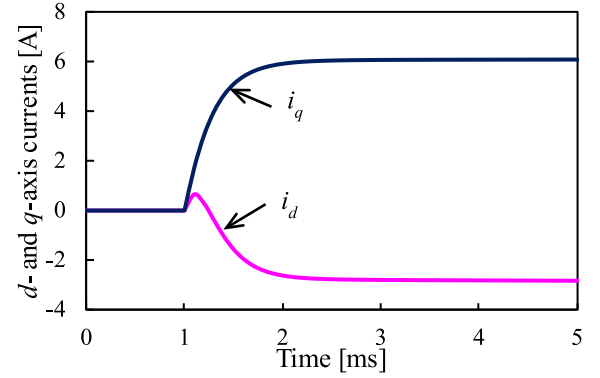
(a) M- and T-axis currents



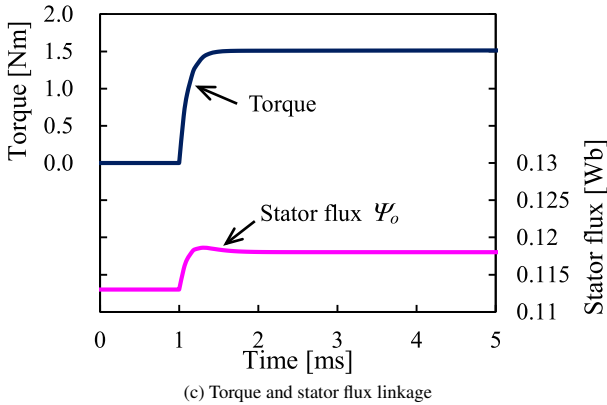
(a) M- and T-axis currents



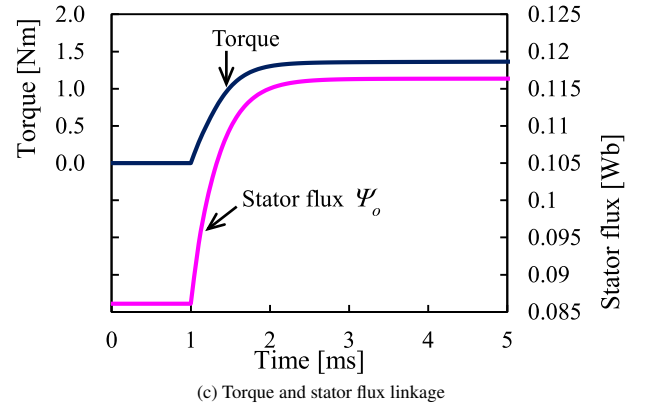
(b) d- and q-axis currents



(b) d- and q-axis currents



(c) Torque and stator flux linkage



(c) Torque and stator flux linkage

Fig. 10. Current step response (SPMSM, simulation)

Fig. 11. Current step response (IPMSM, simulation)

rated armature current of 6.79 A. Rotor speed is 2000 min⁻¹.

As shown in Fig. 10, which is results of SPMSM, M- and T-axis currents follows the reference values, as well as the conventional d- and q-axis current control system. Fig. 10(b) also verified that the $i_d = 0$ control is applied. Torque and stator flux linkage increase as the current changes.

The same is true for IPMSM. Fig. 11 verified that the proposed method for the reference current calculation given by (16) achieves the MTPA control.

4.3 Experimental Results Experiments were performed to confirm the effectiveness of the proposed method. The control period of the current control system is 100 μ s. An insulated gate bipolar transistor module is used for the inverter. The pulse width modulation (PWM) carrier frequency is 10 kHz. The dead-time of the inverter is 3.5 μ s. The DC link voltage of the inverter is 200 V.

The tested motor is connected to an induction machine (IM) used as a load. The rotor speed is controlled by a

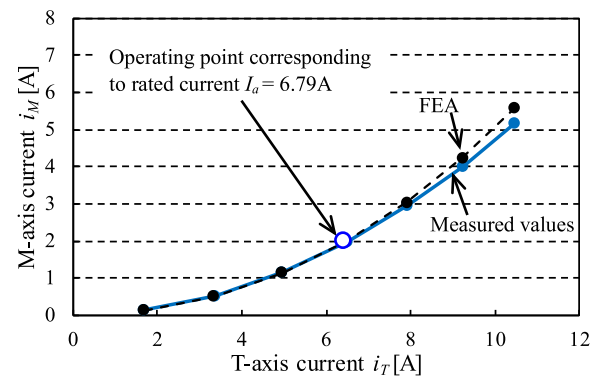


Fig. 12. M- and T-axis currents characteristic of the tested SPMSM

load-side IM. In this paper, the rotor speed is 2000 min⁻¹.

Fig. 12 shows experimental result of M- and T-axis currents characteristic of the tested SPMSM. In experiment, the

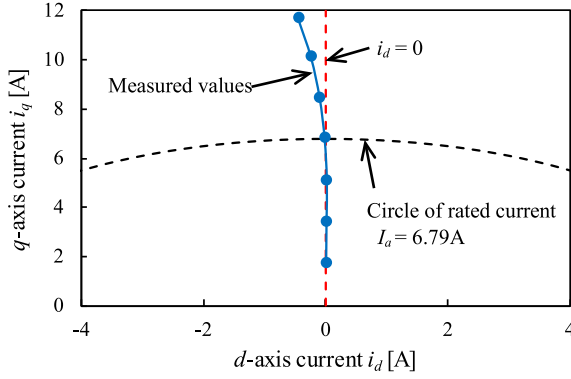


Fig. 13. d - and q -axis current trajectory of the tested SPMSM when proposed method is applied

coefficient k_{iM-iT} is 0.0474 A^{-1} , as well as simulation. The experimental result agrees with the FEA below the rated current. Characteristic differences occur in the operating conditions exceeding the rated current because the value of k_{iM-iT} is given a constant corresponding to the rated current.

Fig. 13 shows d - and q -axis current trajectory of the tested SPMSM when the proposed method is applied. The experimental result shown in Fig. 13 also verified that the proposed method for the reference current calculation given by (16) achieves the MTPA control.

In Fig. 13, the negative d -axis current exceeding the rated current is likely due to a modeling error in the proposed model. Assigning the rated current value as a constant, k_{iM-iT} , in (16) is a probable cause.

5. Conclusions

In this paper, the relationship between i_M and i_T under the MTPA condition of PMSM was expressed by using machine parameters. Moreover, using this relationship, M- and T-axis currents vector control is proposed. It was confirmed that the relationship between i_M and i_T can be approximated by a quadratic function for SPMSM and IPMSM. In addition, approximately accurate results are obtained by the proposed control method in both cases.

However, modeling errors occur due to overload, improvement of the proposed model is required in the future. In addition, applying the proposed model to PMSM drive systems equipped with torque control is a subject for future study.

Acknowledgment

The authors would like to thank Yu Kaieda and Daiki Sekiguchi for their help in conducting this study.

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